GOVERNMENT OF KARNATAKA

KARNATAKA STATE EXAMINATION AND ASSESSMENT BOARD

2026 II PUC EXAM – 1

SCHEME OF EVALUATION

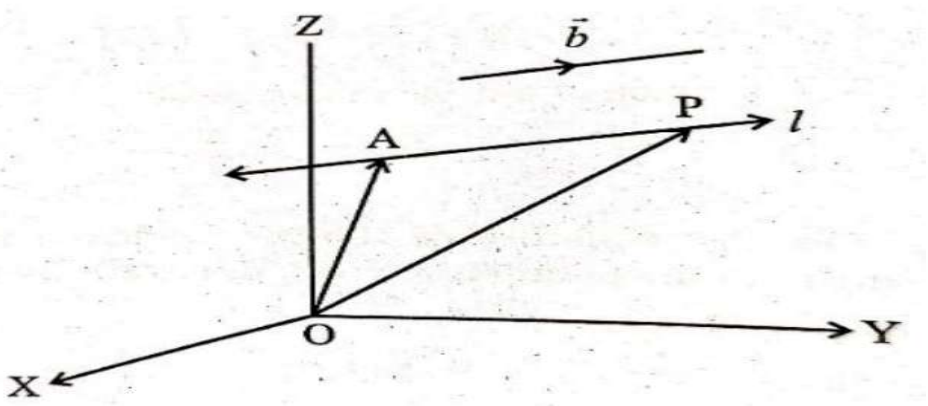
SUBJECT: 35 – MATHEMATICS **MAX. MARKS: 80**

Instructions: a) Any answer by alternate method should be valued and suitably awarded.
b) All answers (including extra , struck off and repeated) should be valued. Answers with maximum marks must be considered .

Q NO:	PART-A	MARKS
I 1	c) or symmetric and transitive	1
2	d) or $(-\infty,\infty)$	1
3	b) or 2	1
4	a) or 25	1
5	b) or $(-1, 1)$	1
6	c) or 1	1
7	a) or 0	1
8	d) or Statement I is false and Statement II is true	1
9	c) or $-cosec^{-1} x + c$	1
10	b) or 0	1
11	d) or $a = \frac{1}{\lvert\lambda\rvert}$	1
12	a) or $3\hat{i} + 2\hat{j} + \hat{k}$	1
13	c) or K , 0 , 0	1
14	a) or $\frac{1}{36}$	1
15	b) or 0.12	1
II 16	- 1	1
17	0	1
18	3	1
19	2	1
20	1	1

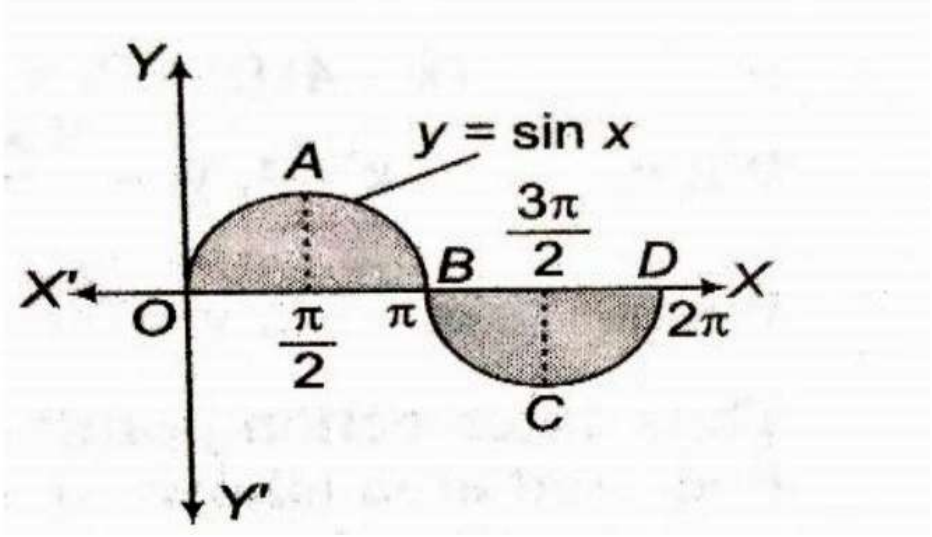
	PART-B	
21	Writing $\tan^{-1}\sqrt{\frac{1-\cos x}{1+\cos x}} = \tan^{-1}\sqrt{\frac{2sin^2\frac{x}{2}}{2cos^2\frac{x}{2}}}$	1
	Getting $\tan^{-1}(\tan\frac{x}{2}) = \frac{x}{2}$	1
22	Writing area of triangle $\Delta ABP = \frac{1}{2}\begin{vmatrix} 1 & 2 & 1 \\ 3 & 6 & 1 \\ x & y & 1 \end{vmatrix} = 0$	1
	Getting $2y - 4x = 0$ OR y = 2x as the equation of line	1
23	Writing $\frac{dy}{dx} = -2\sin 2t$ and $\frac{dx}{dt} = \cos t$	1
	Getting $\frac{dy}{dx} = \frac{-2\sin 2t}{\cos t}$	1
24	Writing $C = 2\pi r$	1
	Getting $\frac{dC}{dt} = 2\pi\frac{dr}{dt} = 2\pi$ X0.7 $= 1.4\pi$ cm/s	1
25	Writing G I $= \int \frac{1}{x(1+\log x)}\,dx$ OR put $1 + \log x = t$ and $\frac{1}{x}dx = dt$	1
	Getting $I = \log\lvert 1 + \log x\rvert + c$	1
26	Writing $\int \frac{dy}{1+y^2} = \int \frac{dx}{1+x^2}$	1
	Getting $\tan^{-1} y = \tan^{-1} x + c$	1
27	Getting $\vec{a}\cdot\vec{b} = 10$	1
	Required $= \frac{\vec{a}\cdot\vec{b}}{\lvert\vec{b}\rvert} = \frac{10}{\sqrt{6}}$	1
28	*Getting* $\overrightarrow{b_1}\cdot\overrightarrow{b_2} = 18$ OR $\lvert\overrightarrow{b_1}\rvert = 9$ and $\lvert\overrightarrow{b_2}\rvert = 3$	1
	Getting $\cos\theta = \frac{\overrightarrow{b_1}\cdot\overrightarrow{b_2}}{\lvert\overrightarrow{b_1}\rvert\lvert\overrightarrow{b_2}\rvert} = \frac{2}{3}$ and $\theta = \cos^{-1}(\frac{2}{3})$	1
29	Writing S = { (b,b),(b,g),(g,b),(g,g)} And E = { (b,b) and F = {(b,b),(b,g),(g,b)}	1
	Writing E ∩ F = { (b,b) } and getting $P(\frac{E}{F}) = \frac{P(E\cap F)}{P(F)} = \frac{1}{3}$	1

	PART-C	
30	Every triangle is congruent to itself . $(T, T) \in R \ \forall \ T \in T$ $\therefore$ R is reflexive	1
	Let $(T_1, T_2) \in R$ => T_1 is congruent to T_2 => T_2 is congruent to T_1 $(T_2, T_1) \in R$ $\therefore$ R is symmetric	1
	Let $(T_1, T_2) \in R$ and $(T_2, T_3) \in R$ => T_1 is congruent to T_2 and T_2 is congruent to T_3 => T_1 is congruent to T_3 $(T_1, T_3) \in R$ $\therefore$ R is transitive => R is an equivalence relation .	1
31	Put $x = \cos\theta$ => $\theta = \cos^{-1} x$	1
	Getting RHS $= cos^{-1}(4\cos^3\theta - 3\cos\theta) = \cos^{-1}(cos\ 3\theta)$	1
	RHS $= 3\theta = 3\cos^{-1} x$ = LHS	1
32	*Writing* $P = \frac{A + A^1}{2} = \begin{bmatrix} 1 & 2 \\ 2 & 2 \end{bmatrix}$ is symmetric	1
	Writing $Q = \frac{A - A^1}{2} = \begin{bmatrix} 0 & 3 \\ -3 & 0 \end{bmatrix}$ is skew- symmetric	1
	Getting $P + Q = \begin{bmatrix} 1 & 5 \\ -1 & 2 \end{bmatrix} = A$	1
33	Writing $log\ y = sin\ x\ log\ x$	1
	Getting $\frac{1}{y}\frac{dy}{dx} = \frac{sin\ x}{x} + \log x \,.\, \cos x$	1
	Getting $\frac{dy}{dx} = x^{sin\ x}\left(\frac{sin\ x}{x} + \log x \,.\, \cos x\right)$	1
34	Writing $f^{\prime}(x) = 6x^2 - 30x + 36$ *OR* *taking* $f^{\prime}(x) = 0$ *and getting* $x = 2$ *and* $x = 3$	1
	Writing $f(1) = 24$, $f(2) = 29$ and $f(3) = 28$ and $f(5) = 56$	1
	writing absolute Maximum value of f is 56 occurs at at $x = 5$ and absolute Minimum value of f is 24 occurs at at $x = 1$	1

35	Writing $\frac{1}{(x+1)(x+2)} = \frac{A}{(x+1)} + \frac{B}{(x+2)}$	1
	getting $A = 1$, $B = -1$	1
	Getting GI $= \int \frac{1}{x+1}\, dx + \int \frac{-1}{x+2}\, dx$ $= \log \lvert x+1 \rvert - \log \lvert x+2 \rvert + c$ OR $GI = \log \left\lvert \frac{x+1}{x+2} \right\rvert + C$	1
36	Writing $\overrightarrow{AB} = \hat{j} + 2\hat{k}$ and $\overrightarrow{AC} = \hat{i} + 2\hat{j}$	1
	Getting $\overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 1 & 2 \\ 1 & 2 & 0 \end{vmatrix} = -4\hat{i} + 2\hat{j} - \hat{k}$	1
	Getting Area $= \frac{1}{2} \lvert \overrightarrow{AB} \times \overrightarrow{AC} \rvert = \frac{\sqrt{21}}{2}$	1
37	 Correct figure award 1 mark (Note : wrong figure deduct 1 mark)	1
	writing $\overrightarrow{AP} \parallel \vec{b}$ *and* $\overrightarrow{AP} = \lambda \vec{b}$	1
	Getting $\vec{r} = \vec{a} + \lambda \vec{b}$	1
38	Let E_1 and E_2 be the events choosing Bag I and Bag II and A be the event of drawing red ball $P(E_1) = P(E_2) = \frac{1}{2}$	1
	getting $P(A\lvert E_1) = \frac{3}{7}$ and $P(A\lvert E_2) = \frac{5}{11}$	1
	Getting $P(E_2\lvert A) = \frac{P(A\lvert E_2)\cdot P(E_2)}{P(A\lvert E_1)\cdot P(E_1) + P(A\lvert E_2)\cdot P(E_2)} = \frac{35}{68}$	1

	PART-D	
39	**Define g : R → R by g(y) = $\frac{y-3}{4}$**	1
	Proving gof (x) = x => gof = I_R	1
	Proving fog (y) = xy => fog = I_R	1
	gof = fog = I_R and Writing f is invertible	1
	Writing $f^{-1} = g$ OR $f^{-1}(x) = \frac{x-3}{4}$ OR $f^{-1}(y) = \frac{y-3}{4}$	1
	OR	
	Let $f(x_1) = f(x_2)$ $x_1, x_2 \in A$ => $4x_1 + 3 = 4x_2 + 3$	1
	=> $x_1 = x_2$ => f is one - one	1
	$f(x) = y = 4x + 3$ => $x = \frac{y-3}{4} \in R$	1
	$\forall$ $y \in R$ *there exists* $x = \frac{y-3}{4} \in R$ such that f(x) = y => f is onto	1
	$\therefore$ f is invertible and $f^{-1}(x) = \frac{x-3}{4}$ OR $f^{-1}(y) = \frac{y-3}{4}$	1
40	**Getting $AB = \begin{bmatrix} -1 & 2 & 1 \\ 4 & -8 & -4 \\ -3 & 6 & 3 \end{bmatrix}$**	1
	Getting $(AB)^1 = \begin{bmatrix} -1 & 4 & -3 \\ 2 & -8 & 6 \\ 1 & -4 & 3 \end{bmatrix}$	1
	Getting $B^1 = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}$ and A^1 = [1 -4 3]	1
	Getting $B^1A^1 = \begin{bmatrix} -1 & 4 & -3 \\ 2 & -8 & 6 \\ 1 & -4 & 3 \end{bmatrix}$	1
	Writing $(AB)^1 = B^1 A^1$	1

41	$A=\begin{bmatrix}3 & -2 & 3\\ 2 & 1 & -1\\ 4 & -3 & 2\end{bmatrix}, X=\begin{bmatrix}x\\ y\\ z\end{bmatrix}, B=\begin{bmatrix}8\\ 1\\ 4\end{bmatrix}$ OR Getting $\lvert A\rvert=-17$	1
	Getting adj A $=\begin{bmatrix}-1 & -5 & -1\\ -8 & -6 & 9\\ -10 & 1 & 7\end{bmatrix}$ NOTE: If any 4 cofactors are correct award 1 mark	2
	Writing $X=A^{-1}B=\frac{adj\,A}{\lvert A\rvert}B=\frac{1}{-17}\begin{bmatrix}-1 & -5 & -1\\ -8 & -6 & 9\\ -10 & 1 & 7\end{bmatrix}\begin{bmatrix}8\\ 1\\ 4\end{bmatrix}$	1
	Getting $x=1, y=2$ and $z=3$	1
42	Getting $\frac{dy}{dx}=Ame^{mx}+Bne^{nx}$	1
	Getting $\frac{d^2y}{dx^2}=Am^2e^{mx}+Bn^2e^{nx}$	1
	Writing LHS = $Am^2e^{mx}+Bn^2e^{nx}$ – (m + n) $(Ame^{mx}+Bne^{nx})$ +mn $(Ae^{mx}+Be^{nx})$	1
	Getting LHS = $Am^2e^{mx}+Bn^2e^{nx}-Am^2e^{mx}-Bmne^{nx}-Amne^{mx}-Bn^2e^{nx}+A\,mne^{mx}+Bmne^{nx}$	1
	Showing LHS = 0 = RHS OR $\frac{d^2y}{dx^2}-(m+n)\frac{dy}{dx}+mn\,y=0$	1
43	Writing I = $\frac{1}{2a}\int\frac{(a+x)+(a-x)}{(a+x)(a-x)}\,dx$	1
	Getting I = $\frac{1}{2a}\int(\frac{1}{a-x}+\frac{1}{a+x})\,dx$	1
	Getting I = $\frac{1}{2a}$ [- log | a – x | + log | a + x |]	1
	Getting I = $\frac{1}{2a}\log\left\lvert\frac{a+x}{a-x}\right\rvert$ + c	1
	Getting $\int\frac{1}{25-x^2}\,dx=\frac{1}{10}log\left\lvert\frac{5+x}{5-x}\right\rvert$ + c	1

44		1
	Drawing correct figure (for wrong figure deduct 1 mark)	
	Required Area = $2\int_0^{\pi} sin\ x\ dx$	1
	$= 2\,(-\cos x)_0^{\pi}$	1
	$= -2\,(\cos\pi - \cos 0)$	1
	Required area = $-2\,(-1-1) = 4$ sq units	1
	OR	
	For Figure	1
	Required Area = $4\int_0^{\frac{\pi}{2}} sin\ x\ dx$	1
	$= 4\,(-\cos x)_0^{\frac{\pi}{2}}$	1
	$= -4\,(\cos\frac{\pi}{2} - \cos 0)$	1
	$= -4\,(-1) = 4$	1
45	Writing $\frac{dy}{dx} + \frac{2}{x}y = x$	1
	$P = \frac{2}{x}$ and $Q = x$	1
	$IF = e^{\int p\,dx} = e^{\int \frac{2}{x}dx} = x^2$	1
	GS is $y\,(IF) = \int Q.(IF)dx + c$ OR $y.x^2 = \int x^3\,dx + c$	1
	Getting $y\,x^2 = \frac{x^4}{4} + c$	1

PART – E

46

RHS = $\int_0^a f(a-x)\,dx$

Put a – x = t => dx = - dt and x = 0 , t = a

x = a , t = 0 — **1**

Getting RHS = $\int_a^0 f(t)(-dt)$ — **1**

Getting RHS = $\int_0^a f(x)\,dx$ = LHS — **1**

Let I = $\int_0^{\frac{\pi}{2}} \frac{\sqrt{\sin x}}{\sqrt{\sin x}+\sqrt{\cos x}}\,dx$ ---------(1) — **1**

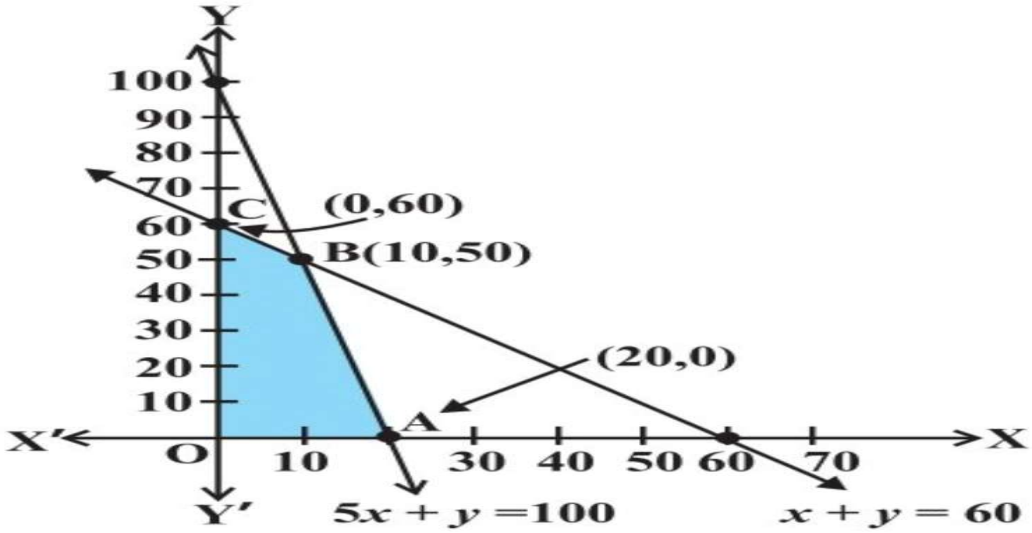
Changing x to $\frac{\pi}{2}-x$

Getting I = $\int_0^{\frac{\pi}{2}} \frac{\sqrt{\cos x}}{\sqrt{\cos x}+\sqrt{\sin x}}\,dx$ ---------(2)

Adding (1) and (2) => $2I = \int_0^{\frac{\pi}{2}} 1\,dx$ — **1**

Getting I = $\frac{\pi}{4}$ — **1**

OR



1+1+1

Drawing line x + y = 60 correctly award 1 mark

Drawing line 25x + 5y = 500 or 5x +y = 100 correctly award 1 mark

For Shading the feasible region award 1 mark

Coner points	Z = 250x + 75y
O(0 , 0)	0
A(0 , 60)	4500
E(10 , 50)	6250
D(20 , 0)	5000

1 + 1

Maximum value of Z is 6250 occurs at the point (10 , 50) — **1**

47	**Getting** $A^2 = \begin{bmatrix} 7 & 12 \\ 4 & 7 \end{bmatrix}$	1
	Showing $A^2 - 4A + I = O$	1
	Getting $A - 4I + A^{-1} = O$ **OR** $A^{-1} = 4I - A$	1
	Getting $A^{-1} = \begin{bmatrix} 2 & -3 \\ -1 & 2 \end{bmatrix}$	1
	OR	
	Writing At $x = 2$ **LHL =RHL** $5 = 2a + b$ -------(1)	1
	At $x = 10$ **LHL =RHL** $10\,a + b = 21$ -------(2)	1
	Getting $a = 2$	1
	Getting $b = 1$	1